

Bank Account Management System using Classes and Objects

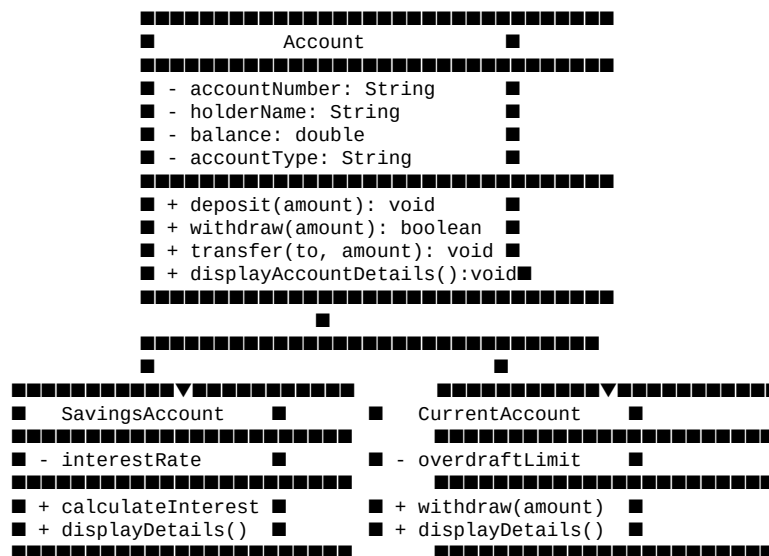
Objective / Aim

To design and implement a Bank Account Management System in Java using classes and objects, demonstrating attributes, methods, object creation, deposits, withdrawals, interest calculation, and fund transfers.

Algorithm / Logic Description

1. Create a base class named Account.
2. Define account number, holder name, balance, and account type.
3. Create SavingsAccount with interestRate.
4. Create CurrentAccount with overdraftLimit.
5. Define deposit(), withdraw(), transfer(), and displayAccountDetails().
6. Define calculateInterest() in SavingsAccount.
7. Override withdraw() in CurrentAccount for overdraft support.
8. Create savings and current account objects.
9. Perform deposits, withdrawals, transfers, and interest calculations.
10. Display updated account details and transaction results.

UML Class Diagram



Relationship: SavingsAccount and CurrentAccount inherit from Account.

Source Code

```

class Account {
    protected String accountNumber, holderName, accountType;
    protected double balance;

    Account(String accountNumber, String holderName,
            double balance, String accountType) {
        this.accountNumber = accountNumber;
        this.holderName = holderName;
        this.balance = balance;
        this.accountType = accountType;
    }

    public void deposit(double amount) {
        if (amount > 0) balance += amount;
    }

    public boolean withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            return true;
        }
        return false;
    }

    public void transfer(Account target, double amount) {
        if (withdraw(amount)) target.deposit(amount);
    }

    public void displayAccountDetails() {
  
```

```

        System.out.println("Account Number : " + accountNumber);
        System.out.println("Holder Name : " + holderName);
        System.out.println("Account Type : " + accountType);
        System.out.printf("Balance : Rs.%.2f%n", balance);
    }
}

class SavingsAccount extends Account {
    private double interestRate;

    SavingsAccount(String no, String name, double balance, double rate) {
        super(no, name, balance, "Savings");
        interestRate = rate;
    }

    public double calculateInterest() {
        return balance * interestRate / 100;
    }

    @Override
    public void displayAccountDetails() {
        super.displayAccountDetails();
        System.out.println("Interest Rate : " + interestRate + "%");
        System.out.printf("Interest : Rs.%.2f%n", calculateInterest());
    }
}

class CurrentAccount extends Account {
    private double overdraftLimit;

    CurrentAccount(String no, String name, double balance, double limit) {
        super(no, name, balance, "Current");
        overdraftLimit = limit;
    }

    @Override
    public boolean withdraw(double amount) {
        if (amount > 0 && amount <= balance + overdraftLimit) {
            balance -= amount;
            return true;
        }
        return false;
    }

    @Override
    public void displayAccountDetails() {
        super.displayAccountDetails();
        System.out.printf("Overdraft Limit: Rs.%.2f%n", overdraftLimit);
    }
}

public class BankAccountDemo {
    public static void main(String[] args) {
        SavingsAccount savings =
            new SavingsAccount("SA205", "Arjun", 12000, 6.0);
        CurrentAccount current =
            new CurrentAccount("CA305", "Meera", 18000, 6000);

        savings.deposit(3000);
        savings.withdraw(2000);
        current.withdraw(20000);
        savings.transfer(current, 2500);

        savings.displayAccountDetails();
        current.displayAccountDetails();
    }
}

```

Compilation & Execution Commands

```
javac BankAccountDemo.java
java BankAccountDemo
```

Output Screen — New Sample

```
===== SAVINGS ACCOUNT =====
Account Number : SA205
Holder Name    : Arjun
Account Type   : Savings
Balance       : Rs.12000.00
Interest Rate  : 6.0%
Interest      : Rs.720.00

Depositing Rs.3000...
Deposited: Rs.3000.0

Withdrawing Rs.2000...
Withdrawn: Rs.2000.0

===== CURRENT ACCOUNT =====
Account Number : CA305
Holder Name    : Meera
Account Type   : Current
Balance       : Rs.18000.00
Overdraft Limit: Rs.6000.00

Withdrawing Rs.20000...
Withdrawn: Rs.20000.0

===== TRANSFER =====
Transferring Rs.2500 from Savings to Current...
Withdrawn: Rs.2500.0
Deposited: Rs.2500.0
Transferred: Rs.2500.0 to CA305

===== FINAL ACCOUNT DETAILS =====
Savings Account:
Account Number : SA205
Holder Name    : Arjun
Account Type   : Savings
Balance       : Rs.10500.00
Interest Rate  : 6.0%
Interest      : Rs.630.00

Current Account:
Account Number : CA305
Holder Name    : Meera
Account Type   : Current
Balance       : Rs.500.00
Overdraft Limit: Rs.6000.00
```

Result

The Bank Account Management System was successfully implemented in Java using classes and objects. The program demonstrates savings and current account objects, deposits, withdrawals, interest calculation, overdraft support, and fund transfers using a new sample output.